AI Assignment

Release: 28-September-2026 **Deadline**: 31-October-2026

1 Overview

In this assignment, you will develop a **general-purpose AI reasoning system** that determines whether a given statement is logically entailed by a knowledge base containing facts and rules.
The central task is:

Natural Language Knowledge Base → Internal Representation → Classical Reasoning → True/False

You are free to choose how you convert the natural-language input into an internal representation. However, the **final reasoning process must be performed using classical AI reasoning techniques**.

2 Problem Statement

You are given a knowledge base consisting of:

- facts that describe known information,
- rules that describe relationships between facts, and
- a query whose logical status must be determined.

For each query, your system must determine whether the query is **entailed** by the knowledge base.
For example:

Bob is young.
Bob is kind.
If someone is young, then they are smart.
If someone is smart and kind, then they are successful.

Query:

Bob is successful.

The correct answer is:

`True`

because the system can derive:

$$Young(Bob) \rightarrow Smart(Bob)$$

and

$$Smart(Bob) \land Kind(Bob) \rightarrow Successful(Bob).$$

You must develop an end-to-end reasoning system that takes a knowledge base (**context**) and a query (**statement**) as input and produces a prediction (**flag**):

`True` or `False`.

3 Dataset

For the development of your reasoning system, you must use the **RuleTaker dataset**, **CounterBench** and **FOLIO** available on Hugging Face . The dataset contains knowledge bases consisting of facts and rules, together with statements for which the system must determine whether they are logically entailed. The final assignment will be evaluated on a **private test dataset** containing previously unseen knowledge bases and queries. The private test set will not be released to students. The goal is therefore to build a **general reasoning system**, rather than a system that memorizes the public examples.

4 Reasoning Algorithm Requirement

The final inference must be performed using **classical AI reasoning techniques**. You may use and combine techniques covered in the course, including: Forward Chaining, Backward Chaining, Unification, Resolution, Breadth-First Search (BFS), Depth-First Search (DFS), Uniform-Cost Search (UCS), A* or other heuristic search methods, other classical reasoning techniques discussed in the course.
You are **not required to implement every algorithm**. You should choose an appropriate strategy or combination of strategies for your reasoning system.

The conversion from natural language into your internal representation is considered part of your system design. You may use any appropriate computational method for this conversion. However, the final decision must be obtained through your own classical reasoning engine.
For example, the following pipeline is acceptable:

$$\text{Natural Language} \rightarrow \text{Parser / Representation} \rightarrow \text{Facts + Rules} \rightarrow \text{Classical Reasoning} \rightarrow \text{Prediction.}$$

The following approach is **not permitted**:

$$\text{Natural Language} \rightarrow \text{Generative AI} \rightarrow \texttt{True/False}.$$

5 Evaluation

The assignment will primarily evaluate the system on the **private test set**. For every query, your system should produce the predicted label (`True` or `False`).

Reasoning Accuracy The primary evaluation metric is prediction accuracy:

$$\textit{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Queries}} \times 100.$$

Reasoning Efficiency Efficiency will be evaluated using the computational effort required by the reasoning engine i.e total execution time.

Rules

1. The objective is to build a **general-purpose reasoning system**, not a system specifically designed for the released dataset.
2. The private test dataset must not be used during development.
3. Students must not manually create predictions for the private test set.
4. Directly using a generative AI system to produce the final answer to a query is not permitted.
5. Use python as the implementation language. Standard programming libraries and tools may be used for implementation and data processing.
6. Code will be checked for plagiarism.

6 Submission

Each individual must submit:

```
submission/
|
|-- reasoning_agent.ipynb
|
|-- report.pdf
|
|-- submission.csv
|
‘-- README.txt
```

The notebook must contain the complete implementation of the reasoning system with final accuracy and inference time printed and it should be able to generate a submission.csv file which contains id and predicted labels.
The report should briefly describe:

1. the chosen representation,
2. the reasoning algorithms used,
3. experiments performed on the public dataset,
4. examples of reasoning traces.